```python
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

# Given data
n1 = 30
n2 = 30

mean1 = 1000      # Store A mean sales
mean2 = 950       # Store B mean sales

sd1 = 100         # Store A standard deviation
sd2 = 120         # Store B standard deviation

alpha = 0.05

# Standard Error
se = np.sqrt((sd1**2 / n1) + (sd2**2 / n2))

# t-statistic
t_stat = (mean1 - mean2) / se

# Welch's degrees of freedom
df = ((sd1**2 / n1 + sd2**2 / n2)**2) / (
    ((sd1**2 / n1)**2 / (n1 - 1)) +
    ((sd2**2 / n2)**2 / (n2 - 1))
)

# Two-tailed p-value
p_value = 2 * stats.t.sf(abs(t_stat), df)

# Critical t-value
t_critical = stats.t.ppf(1 - alpha/2, df)

# 95% Confidence Interval
difference = mean1 - mean2
ci_lower = difference - t_critical * se
ci_upper = difference + t_critical * se

# Display Results
print("Independent Two-Sample t-Test")
print("-" * 40)
print(f"Store A Mean Sales      : ${mean1:.2f}")
print(f"Store B Mean Sales      : ${mean2:.2f}")
print(f"Mean Difference         : ${difference:.2f}")
print(f"Standard Error          : {se:.4f}")
print(f"Degrees of Freedom      : {df:.2f}")
print(f"t-statistic             : {t_stat:.4f}")
print(f"Critical t-value        : ±{t_critical:.4f}")
print(f"Two-tailed p-value      : {p_value:.4f}")
print(f"95% Confidence Interval : (${ci_lower:.2f}, ${ci_upper:.2f})")
```

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Independent Two-Sample t-Test
----------------------------------------
Store A Mean Sales      : $1000.00
Store B Mean Sales      : $950.00
Mean Difference         : $50.00
Standard Error          : 28.5190
Degrees of Freedom      : 56.17
t-statistic             : 1.7532
Critical t-value        : ±2.0031
Two-tailed p-value      : 0.0850
95% Confidence Interval : ($-7.13, $107.13)
```

```
# Decision
print("\nDecision:")
if p_value < alpha:
    print("Reject the null hypothesis (H0).")
    print("Conclusion: There is a significant difference in average
daily sales.")
else:
    print("Fail to reject the null hypothesis (H0).")
    print("Conclusion: There is no significant difference in average
daily sales.")
```

```
Decision:
Fail to reject the null hypothesis (H0).
Conclusion: There is no significant difference in average daily sales.
```

```
# Plot
x = np.linspace(-4, 4, 1000)
y = stats.t.pdf(x, df)

plt.figure(figsize=(10,5))
plt.plot(x, y, label=f"t-distribution (df={df:.1f})")

# Rejection regions
plt.fill_between(x, 0, y, where=(x <= -t_critical),
                 color='red', alpha=0.3)

plt.fill_between(x, 0, y, where=(x >= t_critical),
                 color='red', alpha=0.3)

# Observed t
plt.axvline(t_stat, color='blue', linestyle='--',
            linewidth=2, label=f'Observed t = {t_stat:.2f}')

plt.title("Independent Two-Sample t-Test")
plt.xlabel("t-value")
plt.ylabel("Density")
plt.grid(True)
```

```
plt.legend()
plt.show()
```

